

OSCAR DE LA CERDA

Portfolio: oadlc.github.io

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EDUCATION

Yale University, New Haven, CT

May 2028

- *ABET-accredited B.S in Electrical Engineering, Certificate in Programming*
- *Relevant Coursework:* Electromagnetism, Introduction to Computer Engineering, Introduction to Communications and Control, Circuits and Systems Design, Data Structures and Programming, Digital Systems

University of Iceland, Reykjavik, Iceland

June – July 2025

- *Relevant Coursework:* Renewable Energy Technology, Politics and Economics of Renewables

Tom C. Clark High School, San Antonio, Texas

June 2024

- Questbridge National College Match Finalist; AP Scholar with Distinction; National Hispanic Recognition, College Board

TECHNICAL SKILLS

- *Hardware and Electronics:* Schematic Capture, PCB layout & assembly, circuit design, breadboard prototyping, soldering, microcontroller & sensor integration, FPGA/HDL design
- *Test & Debug:* Oscilloscopes, multimeters, board and system-level debugging, closed-loop control tuning
- *Languages & Tools:* C, C++, Verilog, MATLAB, Racket, PowerShell; Altium, KiCad, LTspice, Zemax

EXPERIENCE

Quantum Network Research, Undergraduate Researcher

June 2026 - Now

- Automated FSM (fast steering mirror) sweep sequences with Python across a quad-cell position sensor, mapping boundary conditions and characterizing sensor response across the full field of view
- Designed and implemented a closed-loop control system to stabilize laser pointing on a quad-cell sensor under simulated FSM disturbances, applying characterized sensor boundaries to inform controller design

Yale Undergraduate Aerospace Association, Sub team Lead

November 2024 - Now

- Designed, simulated, and implemented an analog-to-digital converter (ADC) circuit to enable a Teensy microcontroller to process pressure, voltage, and current sensor data during pre-fuel testing
- Developed a complete avionics sensor-suite schematic integrating GPS, IMU, altimeter, MCU, and S-PLIG modules to improve system reliability and flight data acquisition

Yale Bulldogs Racing, Sub team Member

November 2024 – May 2025

- Assembled and soldered a high-voltage PCB for the team's Formula-style electric race car, supporting power-delivery reliability in competitive environments
- Authored technical documentation detailing internal electrical components to ensure compliance with safety and racing regulations

PROJECTS

Two-Axis Ball-Balancing Stabilizer Platform, Independent Project

June – July 2025

- Designed a 3RPS-style two axis stabilizer around a custom ESP32-WROVER_CAM PCB carrier board (schematic capture and layout in KiCad) driving three MF966R servos.
- Built a Python/OpenCV vision pipeline with HSV ball tracking feeding a dual-axis PID controller driven by a law of cosines inverse-kinematics solver
- Debugged firmware level GPIO/PSRAM and PWM channel conflicts during board bring-up, and migrated servo control from HTTP to UDP to eliminate control-loop latency

Piezoelectricity in Iceland, Independent Project

June – July 2025

- Modeled and evaluated piezoelectric energy-harvesting systems (cantilever beams, diaphragm configurations, harvesting eels) on Icelandic ships to offset greenhouse gas emissions